

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The analysis focuses on identifying spending patterns, customer segments, product preferences, and subscription trends to generate actionable insights and support data-driven business decisions.

2. Dataset Summary

- Total Records: 3,900
- Total Features: 18
- Key Features:
 - Customer Demographics: Age, Gender, Location, Subscription Status
 - Purchase Information: Item Purchased, Category, Purchase Amount, Season, Size, Color
 - Shopping Behavior: Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type
 - Missing Values: 37 records in the Review Rating column.

3. Exploratory Data Analysis Using Python

The dataset was first explored and prepared using Python to understand its structure, quality, and key characteristics.

- Data Loading: Imported the dataset into Python using the `pandas` library.
- Initial Data Exploration: Used `df.info()` to examine the dataset structure, data types, and missing values.
- Statistical Analysis: Used `df.describe()` to generate summary statistics and understand the distribution of numerical variables.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Identified null values and imputed the missing values in the **Review Rating** column using the median rating for each product category.
- **Column Standardization:** Renamed all columns using **snake_case** notation to improve readability and maintain consistency.
- **Feature Engineering:**
 - Created an **age_group** column by categorizing customers into different age groups.
 - Created a **purchase_frequency_days** column to represent the frequency of customer purchases in days.
- **Data Consistency Check:** Compared **discount_applied** and **promo_code_used** to identify redundant information. Since the two fields provided overlapping information, **promo_code_used** was removed.
- **Database Integration:** Established a connection between Python and PostgreSQL and loaded the cleaned DataFrame into the database for further SQL-based analysis.

4. Data Analysis Using SQL (Business Transactions)

Structured SQL analysis was performed in **PostgreSQL** to address key business questions and uncover meaningful patterns in customer purchasing behavior.

1. **Revenue by Gender:** Compared total revenue generated by male and female customers.

	gender character varying (20) 🔒	revenue numeric 🔒
1	Female	75191.00
2	Male	157890.00

2. **High-Spending Discount Users:** Identified customers who applied discounts while spending above the overall average purchase amount.

	customer_id integer	purchase_amount numeric (10,2)
1	2	64.00
2	3	73.00
3	4	90.00
4	7	85.00
5	9	97.00
6	12	68.00
7	13	72.00
8	16	81.00
9	20	90.00
10	22	62.00
Total rows: 839		Query complete 00:00:

3. **Top 5 Products by Rating:** Identified the five products with the highest average customer review ratings.

	item_purchased character varying (100)	avg_review_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison:** Compared the average purchase amount between Standard and Express shipping methods.

	shipping_type character varying (50)	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers:** Analyzed differences in average spending and total revenue between subscribed and non-subscribed customers.

	subscription_status character varying (20) 🔒	total_customer bigint 🔒	spend_amount numeric 🔒	total_revenue numeric 🔒
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products:** Identified the five products with the highest percentage of purchases made using discounts.

	item_purchased character varying (100) 🔒	discount_rate numeric 🔒
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation:** Segmented customers into **New, Returning, and Loyal** groups based on their purchase history.

	customer_segment text 🔒	no_of_customers bigint 🔒
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category:** Identified the three most frequently purchased products within each product category.

	item_rank bigint 🔒	item_purchased character varying (100) 🔒	category character varying (50) 🔒	total_customer bigint 🔒
1	1	Jewelry	Accessories	171
2	2	Sunglasses	Accessories	161
3	3	Belt	Accessories	161
4	1	Pants	Clothing	171
5	2	Blouse	Clothing	171
6	3	Shirt	Clothing	169
7	1	Sandals	Footwear	160
8	2	Shoes	Footwear	150
9	3	Sneakers	Footwear	145
10	1	Jacket	Outerwear	163
11	2	Coat	Outerwear	161

9. **Repeat Buyers & Subscriptions:** Examined the relationship between customers with more than five previous purchases and their subscription status.

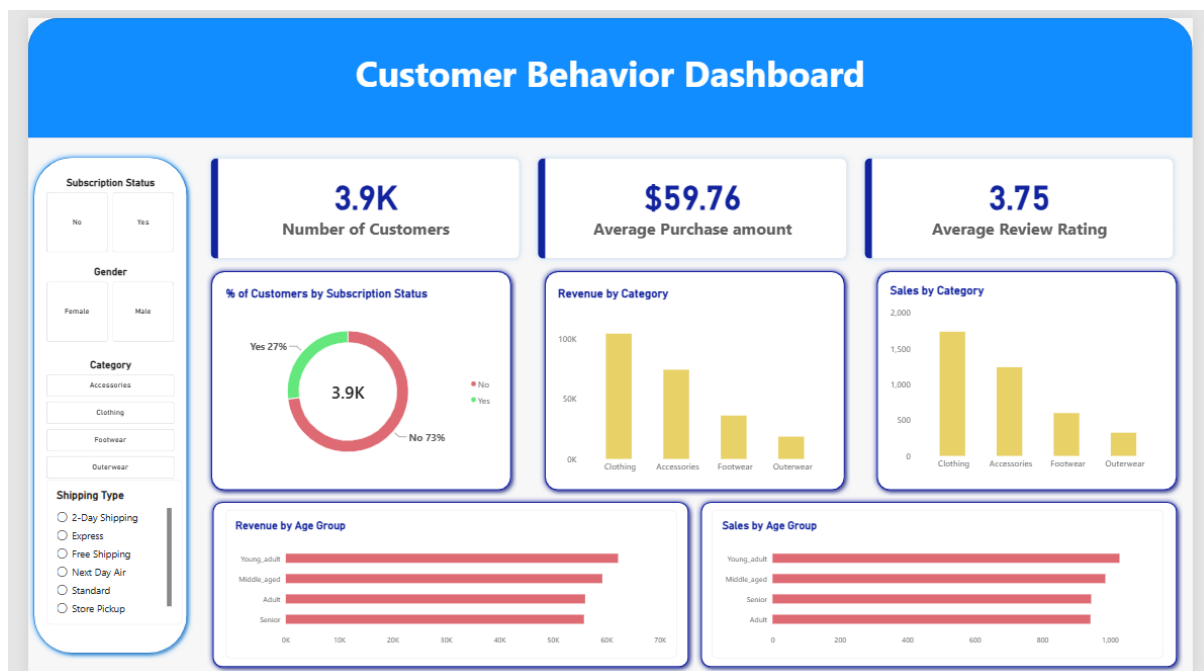
	subscription_status character varying (20) 🔒	repeat_buyers bigint 🔒
1	No	2518
2	Yes	958

10. **Revenue by Age Group:** Analyzed the total revenue contribution of different customer age groups.

	age_group character varying (30) 🔒	total_revenue numeric 🔒
1	Young_adult	62143.00
2	Middle_aged	59197.00
3	Adult	55978.00
4	Senior	55763.00

5. Dashboard Development in Power BI

Finally, an interactive **Power BI dashboard** was developed to visualize the key findings from the analysis and make the insights easier to understand.



6. Business Recommendations

Based on the analysis, the following recommendations can help improve customer engagement, sales performance, and overall business growth:

- **Increase Subscription Adoption:** Promote exclusive benefits, offers, and services to encourage more customers to subscribe.
- **Strengthen Customer Loyalty:** Introduce loyalty programs and rewards for repeat customers to encourage continued purchases and build long-term customer relationships.
- **Optimize Discount Strategy:** Evaluate the impact of discounts on sales and profitability to maintain a balance between increasing purchases and protecting profit margins.
- **Improve Product Positioning:** Feature highly rated and best-selling products prominently in marketing campaigns and promotional activities.
- **Implement Targeted Marketing:** Focus marketing campaigns on high-revenue customer segments and analyze the behavior of customers who prefer express shipping to identify additional opportunities for engagement.